

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Technology Stack Overview

Layer	Technology	Version	Purpose
Frontend / UI	Streamlit	1.32+	Interactive web application
Styling	Custom CSS	-	Dark glassmorphism theme
ML - Deep Learning	TensorFlow / Keras	2.15	BiLSTM model training & inference
ML - Transformers	PyTorch	2.2	BERT model fine-tuning
NLP - Pretrained	HuggingFace Transformers	4.38	bert-base-uncased tokenizer & model
NLP - Utilities	NLTK	3.8	Stopword removal, lemmatization
Generative AI	Google Gemini API	gemini-pro	Personalized guidance generation
Gemini SDK	google-generativeai	0.5+	Python client for Gemini
Data Processing	Pandas	2.2	Dataframe ops, CSV logging
Data Processing	NumPy	1.26	Array/matrix operations
Visualization	Plotly	5.20	Interactive charts (emotion bars,

			trends)
Dataset	Hugging Face `dair-ai/emotion`	-	6-class remapped to 5-class
Environment	Python	3.10+	Core language
Deployment	Streamlit Community Cloud	Free tier	Public web deployment
Version Control	Git + GitHub	-	Source code management
Configuration	python-dotenv	1.0	Environment variable management
Testing	pytest	8.1	Unit testing

Architecture Layers



Why This Stack?

Decision	Rationale
Streamlit over Flask/Django	Rapid development, Python-native, free cloud deployment

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BiLSTM + BERT ensemble	Combines sequential pattern recognition with contextual understanding
Gemini API for guidance	State-of-the-art LLM, generous free tier (1500 req/day), easy Python integration
Gemini-only on cloud	Avoids TensorFlow/PyTorch heavy dependencies on Streamlit's free tier
HuggingFace `dair-ai/emotion`	Publicly available, well-balanced 6-class dataset remappable to our 5 emotions
Plotly for charts	Interactive, aesthetically consistent with dark theme
CSV logging	Zero infrastructure cost, portable, human-readable

Development Tools

Tool	Purpose
VS Code	Primary code editor
GitHub	Version control and repository hosting
Google AI Studio	Gemini API key management
Streamlit Cloud	Free public deployment
pytest	Automated unit testing